

# Mechanics M1 (WME01/01) January 2023 — Recall & Jargon Pack

Tutor

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## Mechanics M1 (WME01/01) January 2023 — Recall & Jargon Pack

**Tutorial:** Student × Tutor · recorded 19 June 2026 · paper practised = **Edexcel IAL Mathematics, Mechanics M1 (WME01/01), January 2023** (log P72070A, 75 marks, 8 questions).

This pack is a fast-recall companion to the full revision guide (guide\_v2/revision\_guide.pdf). It compresses the terms, formulae and methods touched in the lesson into paste-ready tables, and it keeps the **evidence boundaries visible at every line** — you should never have to guess whether a claim comes from the official paper or from the lesson.

**Dual numbering — read both labels.** The printed practice packet used in the lesson was re-ordered relative to the official paper. The collision is *packet Q1 / official Q2*; the train is *packet Q2 / official Q1*; the moments question is *packet Q5 / official Q4*. Every item below carries both labels so the card you worked from matches the official WME01/01 numbering.

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### 0. How to read this pack — provenance & status boundaries

#### 0.1 Where each fact comes from (authority order)

| Source                                                                                                      | Role in this pack                                                                                                                                    | Authority                                                        |
|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| guide_v2/revision_guide.pdf/.tex (this lesson)                                                              | Exact question wording, marks per part, lesson chronology, dual numbering                                                                            | <b>Authoritative</b> — lesson control                            |
| sources/WME01_01_QP_Jan2023 / WME01_01_MS_Jan2023<br>Retained lesson transcript + board/attachment captures | Correct answers, method/accuracy marks, required conditions<br>Lesson chronology, tutor-packet vs official dual numbering, what was actually reached | <b>Authoritative</b> — answers/values<br><b>Primary (lesson)</b> |
| shared/Edexcel_IAL_Mathematics_2018_-<br>specification.txt                                                  | Confirms every topic is in M1 scope                                                                                                                  | Scope truth                                                      |

**Rule applied throughout (official QP/MS vs lesson):** where the lesson narrative's *wording or numbers* disagree with the official QP/MS, **the QP/MS win**. The lesson is used only for *method, diagnosis and technique advice*. Every correction to an earlier guide version is tabulated in §8.

## 0.2 Status legend (used in the “Lesson status” column)

| Tag                                   | Meaning                                                                                                                |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>COMPLETED WITH PROMPTING</b>       | Worked through to the requested answer during the lesson, with tutor prompting                                         |
| <b>REVIEWED ONLY / DEFERRED</b>       | Looked at and commented on, but not re-worked; left for a later day                                                    |
| <b>NOT DISCUSSED</b>                  | Part of the paper, but not opened in the lesson → <b>MS reference only</b>                                             |
| <b>LIVE SET-UP ONLY</b>               | Only the diagram and governing equations were reached live; the arithmetic was not finished → <b>MS reference only</b> |
| <b>ESAT-STYLE WARM-UP — NOT WME01</b> | Opening inequality exercise in the ESAT style; not an M1 / WME01 question                                              |

## 0.3 Source-limited / corrected-guide claims (labelled, not asserted as fact)

Earlier guide versions contained four unsupported claims. All four have been **removed as assertions** and survive only inside explicit “Correction to earlier guide” boxes (quoted in the audit). They are summarised here, flagged, not trusted:

- **C1 — Collision “brief live slip”** ( $x = 2.75$ ,  $v_B = 3.75$ ). The recording and final board show only the correct values  $x = \frac{1}{2}$ ,  $v_B = \frac{3}{2}$ . **No slip occurred**. The claim is removed.
- **C2 — Moments “took moments about B”**. The recording and board show the Student explicitly chose **D**, not B. The actual issue was a signed-equation distance error, corrected to equation (B).
- **C3 — “Q8 method complete”**. Q8 reached only the force diagram + two governing equations live; all numerics are **MS reference only**.
- **C4 — Universal friction rule (“friction  $\leq$  applied pull”)**. Not a valid general rule; the direction of acceleration is decided from the net resolved force. Labelled and not carried forward.

Cells whose numbers come only from the mark scheme (not the lesson) are marked **[MS-only]**; cells carrying a corrected-guide narrative are marked **[Correction]**.

## 1. Recall tables — terms & jargon by topic

Columns are fixed: **Term** • **Student-friendly meaning** • **Exact mathematical meaning** • **Symbols/conditions** • **What it is NOT** • **Lesson example** • **Common trap**. Every “Lesson example” is anchored to this January 2023 paper (dual-labelled); every “Common trap” is from the lesson diagnostic unless marked [MS-only].

### 1.1 Momentum & impulse / direct collision (Packet Q1 / Official WME01 Q2)

| Term                            | Student-friendly meaning                                    | Exact mathematical meaning                                                       | Symbols/conditions                                            | What it is NOT                                            | Lesson example                                                       | Common trap                                                                |
|---------------------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------------|
| Conservation of linear momentum | Total “mass × velocity” before = after, for a closed system | $\sum m_i u_i = \sum m_i v_i$ along the line of impact                           | Velocities are <b>signed</b> (pick a positive direction once) | Not “equal speeds”; not energy conservation               | Q1: $(3m)(1.5) + m(-1.5) = 3mx + m(x+1) \Rightarrow x = \frac{1}{2}$ | Dropping the sign of B’s initial velocity (−1.5)                           |
| Relative speed after impact     | Speeds of the two particles once they separate              | $v_A = \frac{1}{2} \text{ m s}^{-1}, v_B = x + 1 = \frac{3}{2} \text{ m s}^{-1}$ | State <b>which speed is which</b>                             | Two unlabelled numbers may not fully answer               | Q1(a): label “speed of A” and “speed of B”                           | Two unlabelled numbers lose the communication                              |
| Impulse = change of momentum    | The “kick” a force gives over time                          | $I = m(v - u) = \Delta(mv)$                                                      | Impulse <b>on B</b> uses B’s change only                      | Not the total system momentum (that is conserved, ≈0 net) | Q1(b): $I_B = m(\frac{3}{2} - (-1.5)) = 3m \text{ N s}$              | Taking impulse on the wrong particle                                       |
| Magnitude of impulse            | The size of the impulse, always positive                    | $ I_B  = 3m \text{ N s}$                                                         | “Magnitude” ⇒ drop the modulus bars, give a positive quantity | Not a signed value                                        | Q1(b): answer $3m$                                                   | Leaving modulus bars or a sign                                             |
| [Correction] brief live slip    | —                                                           | —                                                                                | —                                                             | —                                                         | Earlier guide claimed $x = 2.75, v_B = 3.75$                         | <b>Removed</b> — recording shows only $x = \frac{1}{2}, v_B = \frac{3}{2}$ |

### 1.2 Constant-acceleration (SUVAT) / speed-time graphs (Packet Q2 / Official WME01 Q1)

| Term                                     | Student-friendly meaning                      | Exact mathematical meaning                                              | Symbols/conditions                           | What it is NOT                                  | Lesson example                                                   | Common trap                                           |
|------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------|-------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------|
| Speed-time graph: area                   | Distance travelled = area under the graph     | $s = \int v dt = \text{area of the region}$                             | Use the shape (trapezium/triangle/rectangle) | Not the slope (that is acceleration)            | Q2(b): area = distance 4800 m                                    | Using gradient instead of area                        |
| Speed-time graph: gradient               | Acceleration = slope of the graph             | $a = \frac{dv}{dt}$ = gradient of a segment                             | Constant segment ⇒ constant $a$              | Not the area                                    | Q2(c): $a = \frac{20}{T} = \frac{20}{40} = 0.5 \text{ m s}^{-2}$ | Reading gradient off the wrong segment                |
| Trapezium-area formula                   | Quick way to sum a constant-speed + two ramps | $A = \frac{1}{2}(a+b)h$ (parallel sides $a, b$ ; height $h$ )           | $a = 180, b = 3T + 180, h = 20$ here         | Not always splitting into triangles + rectangle | Q2(b): $4800 = \frac{1}{2}(180 + 3T + 180)(20)$                  | Always splitting into triangles (slower, error-prone) |
| “Acceleration is twice the deceleration” | The speeding-up phase is steeper              | Ratio of gradients ⇒ deceleration phase lasts <b>2T</b>                 | Base markers: $T, T + 180, 3T + 180$         | Not both phases equal length                    | Q2(a): sketch peak $20 \text{ m s}^{-1}$                         | Assuming both ramp times equal                        |
| Unit consistency                         | Keep units the same on both sides             | $4.8 \text{ km} = 4800 \text{ m}$ before equating to a metre-based area | Area in m (speed × time)                     | Mixing km with m                                | Q2(b): convert first                                             | Equating km to m directly                             |

| Term                        | Student-friendly meaning                 | Exact mathematical meaning                                | Symbols/conditions          | What it is NOT | Lesson example                                                   | Common trap                                     |
|-----------------------------|------------------------------------------|-----------------------------------------------------------|-----------------------------|----------------|------------------------------------------------------------------|-------------------------------------------------|
| Measurement from transition | Durations measured between speed changes | Constant phase runs to $T + 180$ , not to 180 from origin | Read the axis, not “from 0” | —              | Q2(b)/(c)                                                        | Measuring constant phase from the origin        |
| [Correction] peak label     | —                                        | —                                                         | —                           | —              | Evidence supports only a corrected peak to $20 \text{ m s}^{-1}$ | Do <b>not</b> state a mis-drawn value like “30” |

### 1.3 Vectors / constant acceleration (Packet Q3 / Official WME01 Q3) — [REVIEWED ONLY / DEFERRED]

| Term                       | Student-friendly meaning                         | Exact mathematical meaning                                          | Symbols/conditions                                 | What it is NOT                | Lesson example                                                                            | Common trap                                   |
|----------------------------|--------------------------------------------------|---------------------------------------------------------------------|----------------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------|
| Velocity vector            | Rate of change of position                       | $\mathbf{v} = \frac{d\mathbf{r}}{dt}$                               | Components resolved along $\mathbf{i}, \mathbf{j}$ | Not a scalar speed            | Q3(a) [MS-only]: speed at $t = 2$ is $\sqrt{45} = 3\sqrt{5} \approx 6.7 \text{ m s}^{-1}$ | Forgetting the magnitude uses both components |
| Direction of a vector      | Angle the vector makes with $\mathbf{i}$         | $\theta = \tan^{-1}\left(\frac{v_y}{v_x}\right)$ , quadrant-correct | State units (degrees)                              | Not the angle to $\mathbf{j}$ | Q3(b) [MS-only]: angle $27^\circ$ (or $333^\circ$ )                                       | Wrong reference axis                          |
| Time for a given direction | Solve for $t$ when $\mathbf{v}$ points a set way | Set the ratio of components equal to the target slope               | Keep exact until the final value                   | —                             | Q3(c) [MS-only]: $T = 3.2$                                                                | Rounding before solving                       |

### 1.4 Vehicle SUVAT, P-Q-R (Packet Q4 / Official WME01 Q5) — [REVIEWED ONLY / DEFERRED]

| Term                   | Student-friendly meaning                    | Exact mathematical meaning                                        | Symbols/conditions                                    | What it is NOT           | Lesson example                               | Common trap                        |
|------------------------|---------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------|--------------------------|----------------------------------------------|------------------------------------|
| Show that $x = 3u$     | Establish a given relation from the model   | Manipulate the given position/speed expressions to reach $x = 3u$ | “Show that” $\Rightarrow$ end at the displayed result | Not solving for a number | Q4(a) [MS-only]: result $x = 3u$ established | Over-solving past the given result |
| Initial speed $u$      | The starting speed of the vehicle           | Solve the model for $u$                                           | Use the given geometry/times                          | —                        | Q4(b) [MS-only]: $u = 9 \text{ m s}^{-1}$    | Sign/algebra slip in the set-up    |
| Distance in first 14 s | Area under the speed-time graph to $t = 14$ | Sum the relevant areas                                            | Match the stated interval exactly                     | —                        | Q4(c) [MS-only]: distance = 201 m            | Using the wrong time bound         |

### 1.5 Moments & tilting (Packet Q5 / Official WME01 Q4)

| Term                              | Student-friendly meaning                                         | Exact mathematical meaning                                                                            | Symbols/conditions                                                                   | What it is NOT                                 | Lesson example                                       | Common trap                              |
|-----------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------------------------------------------|------------------------------------------------------|------------------------------------------|
| Principle of moments              | For equilibrium about a pivot, clockwise = anticlockwise         | $\sum M_{\text{clockwise}} = \sum M_{\text{anticlockwise}}$                                           | Take moments about a chosen point                                                    | Not a single signed sum (error-prone)          | Q5: equate moments about C and about D               | One long signed sum → sign errors        |
| “On the point of tilting about C” | The support at C is the pivot; the far support reaction is zero  | At tilt about C $\Rightarrow R_D = 0$                                                                 | A force through the pivot, or a zero reaction, has <b>no moment</b> about that pivot | Not “ $R_C = 0$ ”                              | Q5 Scenario 1: $Wx = 150(1.5 - 0.24) = 189$          | Taking the wrong reaction as zero        |
| “On the point of tilting about D” | The support at D is the pivot; the near support reaction is zero | At tilt about D $\Rightarrow R_C = 0$                                                                 | Same idea, opposite end                                                              | —                                              | [Correction] the Student used <b>D</b> , not B       | Claiming moments were taken about B      |
| Dividing two moment equations     | Cancel the common unknown $W$                                    | $\frac{Wx}{189} = \frac{W(0.9 - x)}{81} \Rightarrow \frac{x}{189} = \frac{0.9 - x}{81} = \frac{7}{3}$ | More efficient than substitution when $W$ cancels                                    | Not always — only when a common factor cancels | Q5: $3x = 7(0.9 - x) \Rightarrow x = 0.63 \text{ m}$ | Substitution when division is cleaner    |
| Back-substitution                 | Put the found $x$ into an earlier equation                       | $W = \frac{189}{0.63} = 300 \text{ N}$                                                                | Start from a correct value                                                           | —                                              | Q5: $W = 300 \text{ N}$                              | Starting from a wrong $W$ (“be careful”) |
| Exact distance $x$                | The position of the weight from C                                | $x = 0.63 \text{ m}$                                                                                  | Both values ( $W$ and $x$ ) not split by the paper                                   | Not two separately-marked sub-values           | Q5: 8 marks, both values                             | Assuming the paper splits the 8 marks    |

## 1.6 Boat pulled by two ropes (Packet Q6 / Official WME01 Q6) — [NOT DISCUSSED]

| Term                 | Student-friendly meaning               | Exact mathematical meaning                           | Symbols/conditions                   | What it is NOT              | Lesson example                                       | Common trap                   |
|----------------------|----------------------------------------|------------------------------------------------------|--------------------------------------|-----------------------------|------------------------------------------------------|-------------------------------|
| Two-rope equilibrium | The two rope tensions + weight balance | Resolve both tensions into components; net force = 0 | Each rope has its own angle $\alpha$ | Not a single resultant only | Q6 [MS-only]: rope angle $\alpha \approx 31.9^\circ$ | Assuming both ropes symmetric |
| Tension $P$          | The magnitude of one rope's pull       | Solve the resolved equations for $P$                 | MS accepts 609 N or better (610 N)   | —                           | Q6 [MS-only]: $P \approx 609 \text{ N}$              | Rounding too early            |

## 1.7 Lift and two boxes (Packet Q7 / Official WME01 Q7) — [NOT DISCUSSED]

| Term          | Student-friendly meaning        | Exact mathematical meaning                     | Symbols/conditions     | What it is NOT   | Lesson example                        | Common trap        |
|---------------|---------------------------------|------------------------------------------------|------------------------|------------------|---------------------------------------|--------------------|
| Combined mass | Total mass being lifted         | Sum of the box masses                          | “P and Q” = both boxes | Not just one box | Q7(a) [MS-only]: combined mass = $5m$ | Forgetting one box |
| Mass of P     | The mass of the upper box alone | Solve the contact-force equations for P's mass | Distinct from Q        | —                | Q7(b) [MS-only]: mass of P = $2m$     | Confusing P and Q  |

## 1.8 Parcel on a rough inclined plane (Packet Q8 / Official WME01 Q8) — [LIVE SET-UP ONLY]

| Term                           | Student-friendly meaning                                                         | Exact mathematical meaning                                                                                      | Symbols/conditions                                               | What it is NOT                                                          | Lesson example                     | Common trap                                               |
|--------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------|------------------------------------|-----------------------------------------------------------|
| Resolve parallel to plane      | Newton's 2nd law along the slope                                                 | $18 \cos 40^\circ - F_R - 2g \sin 30^\circ = 2a$                                                                | Up the plane positive; gravity component $2g \sin 30^\circ$ down | Not "vertically" — resolving direction is <b>perpendicular to plane</b> | Q8(1): live                        | Saying forces are "balanced vertically" (wrong direction) |
| Resolve perpendicular to plane | No acceleration into the plane $\Rightarrow$ reaction balances normal components | $R + 18 \sin 40^\circ = 2g \cos 30^\circ$                                                                       | Corrected form; board line was wrong                             | Not $R - 2g \sin 30^\circ + 18 \sin 40^\circ = 0$ (board error)         | Q8(2): corrected perpendicular eqn | Using the uncorrected board line                          |
| Limiting friction              | Max friction before sliding                                                      | $F_R = F_{\max} = \mu R = 0.3R$                                                                                 | Only at the point of slipping                                    | Not "friction = applied pull" (invalid rule)                            | Q8(2): live                        | Assuming friction $\leq$ applied pull decides direction   |
| Acceleration $a$               | Net resolved force $\div$ mass                                                   | $a \approx 1.18 \text{ m s}^{-2}$ (or 1.2) [MS-only]                                                            | Keep unrounded 1.1838 ... for onward use                         | —                                                                       | Q8(a) [MS-only]                    | Rounding before substitution                              |
| Speed at B                     | Use $v^2 = u^2 + 2as$                                                            | $v \approx 3.98 \text{ m s}^{-1}$ (or 4.0) [MS-only]                                                            | $u = 2, s = 5, a = 1.1838 \dots$                                 | —                                                                       | Q8(b) [MS-only]                    | Using a rounded $a$                                       |
| Does it remain at rest?        | Compare down-slope weight component with max friction (force removed)            | $2g \sin 30^\circ = 9.8 \text{ N} > \mu R = 0.3(16.974) \approx 5.09 \text{ N} \Rightarrow$ <b>no</b> [MS-only] | At rest $R = 2g \cos 30^\circ$                                   | Not "friction holds it" by assumption                                   | Q8(c) [MS-only]                    | Asserting rest without comparing                          |

### 1.9 ESAT-style warm-up (inequality) — [ESAT-STYLE WARM-UP — NOT WME01]

| Term                    | Student-friendly meaning                                   | Exact mathematical meaning                                             | Symbols/conditions                                            | What it is NOT          | Lesson example           | Common trap                              |
|-------------------------|------------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------|-------------------------|--------------------------|------------------------------------------|
| Difference $< 1$        | The size of the gap is under 1                             | $ \sqrt{2n} - 7  < 1 \Rightarrow -1 < \sqrt{2n} - 7 < 1$               | Unfold on a number line                                       | Not " $\sqrt{2n} < 1$ " | Warm-up: modulus chain   | Squaring before isolating the root       |
| Inclusive integer count | Count integers strictly between bounds, endpoints excluded | $9 < n < 16 \Rightarrow \{10, \dots, 15\}$ , count $= 15 - 10 + 1 = 6$ | Strict bounds exclude 9 and 16; <b>+1</b> for inclusive count | Not $16 - 9 = 7$        | Warm-up: answer <b>6</b> | Forgetting the "+1"; including endpoints |

## 2. Formula & result availability (M1)

The official M1 formula booklet states there are **no formulae given for M1** beyond those candidates are expected to know; M1 candidates may also need P1/P2 formulae.

| Availability label                            | Items used in this lesson                                                                                                                                                                                                          |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>M1 required — not additionally printed</b> | Momentum $= mv$ ; impulse $= mv - mu$ ; the five constant-acceleration (SUVAT) formulae; Newton's laws; friction law $F = \mu R$ (and $F \leq \mu R$ ); principle of moments. Expected to be known; not in the M1 formula section. |

| Availability label                               | Items used in this lesson                                                                                                                                                      |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>P1/P2 prerequisite formula</b>                | Area of a trapezium, $A = \frac{1}{2}(a + b)h$ (used for the train graph).                                                                                                     |
| <b>Tutor strategy (technique, not a formula)</b> | Clockwise moments = anticlockwise moments; divide two simultaneous equations to cancel a common unknown; box/label key equations; take magnitudes only when the question asks. |
| <b>Official MS reference completion</b>          | The Q8(a) elimination/arithmetic and the Q8(b),(c) answers — supplied from the mark scheme, not worked live.                                                                   |

**Note on rounding and units:** keep signed quantities until a requested magnitude is taken; keep an unrounded value (e.g.  $a = 1.1838 \dots$ ) for onward substitution and only then write  $a \approx 1.18 \text{ m s}^{-2}$ ; use “=” for exact/calculated expressions and “ $\approx$ ” for rounded ones; give SI units on final mechanics answers.

### 3. Dual-numbering & lesson-status ledger

| Packet | Official | Content                                   | Marks              | Lesson status                                      |
|--------|----------|-------------------------------------------|--------------------|----------------------------------------------------|
| —      | —        | ESAT-style inequality warm-up (not WME01) | —                  | Completed w/ prompting ( <b>ESAT — NOT WME01</b> ) |
| Q1     | Q2       | Direct collision and impulse              | 8 (a:5, b:3)       | COMPLETED WITH PROMPTING                           |
| Q2     | Q1       | Train speed-time graph                    | 10 (a:3, b:5, c:2) | COMPLETED WITH PROMPTING                           |
| Q3     | Q3       | Constant-acceleration vectors             | 10                 | REVIEWED ONLY / DEFERRED                           |
| Q4     | Q5       | Vehicle SUVAT (P–Q–R)                     | 9                  | REVIEWED ONLY / DEFERRED                           |
| Q5     | Q4       | Branch on two supports; moments/tilting   | 8 (both values)    | COMPLETED WITH PROMPTING                           |
| Q6     | Q6       | Boat pulled by two ropes                  | 8                  | NOT DISCUSSED                                      |
| Q7     | Q7       | Lift and two boxes                        | 7                  | NOT DISCUSSED                                      |
| Q8     | Q8       | Parcel on a rough inclined plane          | 15 (a:8, b:3, c:4) | LIVE SET-UP ONLY                                   |

Official marks are the exact WME01/01 January 2023 allocations. The packet and official numbers differ only by the two swaps (packet Q1↔Q2 and packet Q4↔Q5); Q3, Q6, Q7 and Q8 carry the same number in both systems.

### 4. Official-reference appendix — answers not worked in this lesson

These values are from the official WME01/01 January 2023 mark scheme. They identify the source of each question; they do not change the lesson status, and none was produced live.

- **Packet Q3 / official Q3 — vectors [REVIEWED ONLY / DEFERRED]:** (a) speed at  $t = 2$ :  $\sqrt{45} = 3\sqrt{5} \approx 6.7 \text{ m s}^{-1}$ ; (b) angle to  $\mathbf{i}$  at  $t = 2$ :  $27^\circ$  (or  $333^\circ$ ); (c)  $T$  for direction  $2\mathbf{i} - 3\mathbf{j}$ :  $T = 3.2$ .

- **Packet Q4 / official Q5 — vehicle SUVAT (P-Q-R) [REVIEWED ONLY / DEFERRED]:** (a) show  $x = 3u$  (given result established); (b)  $u = 9 \text{ m s}^{-1}$ ; (c) distance in first 14 s: 201 m.
  - **Packet Q6 / official Q6 — boat, two ropes [NOT DISCUSSED]:** rope angle  $\alpha \approx 31.9^\circ$ ; tension  $P \approx 609 \text{ N}$  (MS accepts 610 N or better).
  - **Packet Q7 / official Q7 — lift, two boxes [NOT DISCUSSED]:** (a) combined mass of P and Q:  $5m$ ; (b) mass of P:  $2m$ .
  - **Packet Q8 / official Q8 — parcel on rough incline [LIVE SET-UP ONLY]:** (a)  $a \approx 1.18 \text{ m s}^{-2}$  (or 1.2); (b)  $v \approx 3.98 \text{ m s}^{-1}$  (or 4.0); (c) does **not** remain at rest ( $9.8 \text{ N} > 5.09 \text{ N}$ ).
- 

## 5. Closing study actions (from the lesson)

1. **Priority this week:** complete the Pure 2 baseline paper, then the Pure 3 baseline paper, to build a full course diagnostic — ahead of extra M1 work.
2. A separate, already-completed M1 homework paper (marked) is available to look over if time allows.
3. That earlier M1 paper will be revisited together in mid-July, with more homework time allowed first.

**What was not set as homework.** The lesson did not assign “keep practising moments and inclined-plane resolving” or any drill set as homework. Earlier guide versions presented author-created drills and a “recurring weakness” story as tutor-assigned work; that is not supported by the lesson and has been removed.

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## 6. Independent-release status of this pack

This recall pack is an **internal study aid** built from the independently release-audited **PASS** revision guide `guide_v2/revision_guide.pdf` (SHA-256 `0edeb6db0a327900b6579a8c1b49b29a1d78f9fd83d6f399ce260ab17fd296a4`). Nothing here is exported, published, or claimed as a release.